

RAE LAKES, NT, Canada

WMO: 711650

Lat: 64.115N

Lon: 117.313W

Elev: 223

StdP: 98.67

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.1	-40.5	-46.1	0.0	-41.6	-44.0	0.1	-39.6	10.5	-19.8	9.3	-19.8	0.3	0	0.743

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	26.2	16.3	24.5	15.5	22.7	14.9	17.7	23.7	16.7	22.5	15.8	21.2	3.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.3	19.7	14.3	10.4	18.9	13.3	9.8	18.2	50.8	23.6	47.8	22.5	45.0	21.4	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.1	7.0	-44.7	29.0	2.3	1.6	-46.4	30.2	-47.7	31.2	-49.0	32.1	-50.8	33.3		
			-44.0	19.5	1.8	1.7	-45.3	20.8	-46.4	21.8	-47.4	22.7	-48.7	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.8	-26.4	-23.6	-18.4	-6.8	4.4	13.6	17.0	13.7	7.3	-2.1	-14.5	-23.0
	DBStd	16.62	8.17	7.41	8.28	7.37	5.82	4.50	3.57	3.77	4.36	4.56	6.88	8.53
	HDD10.0	5903	1127	941	881	504	188	16	1	9	103	376	734	1022
	HDD18.3	8483	1386	1174	1139	754	432	152	67	149	331	634	984	1281
	CDD10.0	505	0	0	0	0	14	125	219	125	22	0	0	0
	CDD18.3	43	0	0	0	0	0	11	26	7	0	0	0	0
	CDH23.3	283	0	0	0	0	3	87	150	43	0	0	0	0
	CDH26.7	39	0	0	0	0	0	17	18	4	0	0	0	0
Wind	WSAvg	2.8	2.0	2.0	2.5	2.8	3.0	3.1	3.1	3.4	3.6	3.4	2.4	2.0
Precipitation	PrecAvg	298	13	10	13	10	18	36	48	50	39	27	19	15
	PrecMax	354	25	20	27	16	35	93	95	82	79	42	37	27
	PrecMin	224	5	4	3	4	4	7	25	19	11	11	10	5
	PrecStd	35	5	4	6	4	9	18	17	17	15	8	6	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.2	-4.6	4.3	12.4	22.8	29.0	28.7	26.9	20.2	11.2	0.6	-3.7
		MCWB	-4.0	-5.7	1.0	6.4	12.3	17.4	17.7	16.8	13.6	7.6	-0.2	-4.2
	2%	DB	-7.5	-8.5	0.5	9.0	19.5	25.7	26.6	24.2	17.4	7.3	-1.7	-6.0
		MCWB	-7.7	-9.1	-1.5	4.3	10.9	15.4	16.7	16.0	12.3	5.5	-2.3	-6.5
	5%	DB	-12.5	-10.9	-2.5	6.5	16.3	23.5	25.0	21.9	15.4	5.3	-3.6	-8.5
		MCWB	-12.9	-11.4	-4.0	2.7	9.1	14.4	16.1	15.3	11.3	3.8	-4.1	-8.8
	10%	DB	-15.5	-13.0	-5.5	3.9	13.5	21.3	23.4	19.7	13.4	3.4	-5.7	-11.1
		MCWB	-15.9	-13.4	-6.6	1.0	7.4	13.4	15.4	14.3	10.3	2.3	-6.2	-11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.9	-5.3	1.5	6.6	12.8	18.3	19.6	18.5	14.7	7.6	0.0	-4.0
		MCDB	-4.5	-4.6	4.2	11.8	21.4	25.6	25.9	24.3	18.7	10.2	0.6	-3.7
	2%	WB	-8.8	-8.9	-1.4	4.6	11.3	16.4	18.0	17.1	13.0	5.4	-1.9	-6.3
		MCDB	-8.5	-8.4	0.5	9.0	19.0	23.6	24.1	22.1	16.2	7.0	-1.5	-5.9
	5%	WB	-13.2	-11.2	-3.7	3.0	9.6	15.2	17.2	16.0	11.9	3.8	-3.8	-8.5
		MCDB	-12.8	-10.7	-2.3	6.4	15.9	21.6	23.1	21.3	15.0	4.9	-3.4	-8.2
	10%	WB	-16.0	-13.3	-6.3	1.3	7.8	14.1	16.3	14.9	10.7	2.4	-5.9	-11.0
		MCDB	-15.6	-12.9	-5.2	3.9	13.3	20.2	21.8	19.3	13.4	3.3	-5.4	-10.6
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.6	12.7	13.3	10.7	9.4	8.5	7.3	6.0	4.1	6.7	7.8
		MCDBR	9.2	9.9	14.5	14.2	14.4	12.1	10.9	10.6	9.3	5.8	6.0	8.1
		MCWBR	8.7	9.5	12.3	10.2	7.9	5.2	4.9	4.7	5.1	4.1	5.6	7.7
	5% WB	MCDBR	8.8	9.8	14.0	13.6	14.0	10.7	9.8	9.5	8.8	5.4	5.8	7.8
		MCWBR	8.5	9.4	11.9	9.8	7.9	5.0	5.0	4.9	5.2	4.0	5.5	7.6
Clear-Sky Solar Irradiance	taub		0.166	0.231	0.254	0.295	0.329	0.331	0.364	0.333	0.300	0.256	0.163	0.150
	taud		1.969	2.222	2.263	2.231	2.411	2.496	2.412	2.527	2.580	2.414	1.974	1.939
	Ebn at Noon		456	720	853	883	884	888	841	843	808	689	440	224
	Edh at Noon		25	54	81	107	99	93	99	81	62	45	24	10
All-Sky Solar Radiation	RadAvg		0.21	0.99	2.81	4.98	5.80	6.08	5.27	3.88	2.25	0.78	0.22	0.07
	RadStd		0.01	0.05	0.12	0.25	0.42	0.56	0.28	0.29	0.20	0.09	0.03	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page